# Nantong university Course Design

Design Topic: **Reinforced Concrete Cast-in-Place Floor Slab Design**

Course Title: Reinforced Concrete Structures

Department: School of Civil Engineering and Transportation

Class: Civil Engineering

Student ID : 2333130216

Name : TOHA TAMIM AHAMMED

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# Design Specifications for Reinforced Concrete Cast-in-Place Floor Slabs)

## I. Design Description and Design Documents

### 1. Design purpose and method

The task of this course design is to design a reinforced concrete one-way ribbed slab floor system based on the given design data. The aim is to master the calculation of the bearing capacity of the normal and oblique sections of flexural members, and to become familiar with the key points of ribbed slab floor structure layout; to understand the basic concept of redistribution of internal forces during plastic deformation in reinforced concrete; to master the design methods, load values, and internal force calculations for continuous beams and slabs; to be familiar with the methods of drawing bending moment envelope diagrams and resisting moment diagrams; to master the structural requirements for beam and slab reinforcement; and to be familiar with the various regulations in the calculations. Based on the specific circumstances of this course design, the design includes writing the design calculation formulas for the floor slab, secondary beams, and main beams, including the calculation of internal forces in the slab, secondary beams, and main beams, the calculation of reinforcement for the normal and oblique sections, and the drawing of construction drawings.

### 2. Main content and basic requirements of the course design

(1) The internal forces of the slab and secondary beams are calculated according to the plastic internal force distribution; the internal forces of the main beam are calculated according to the elastic theory, and the bending moment envelope diagram is drawn.

(2) Draw the construction drawings of the floor slab structure.

① Floor plan layout (marking wall and column positioning axis numbers and beam and column positioning dimensions and component numbers).

② Floor slab and reinforcement plan (marking slab thickness, diameter, spacing, numbering and positioning dimensions of steel bars in the slab).

③ Secondary beams and reinforcement drawings (marking the cross-sectional dimensions, geometric dimensions, diameter, number, number, and positioning dimensions of the secondary beams).

④ Main beam and reinforcement diagram (draw the bending moment envelope diagram, resisting moment diagram, and reinforcement diagram of the main beam), (label the main beam section).

Dimensions and geometric dimensions, diameter, number, numbering and positioning dimensions of the reinforcing bars.

⑤ Mark the relevant design specifications on the drawing, such as the concrete strength grade, the type of steel reinforcement, and the thickness of the concrete cover.

(3) Calculation sheet: The calculations must be accurate, the steps must be complete, and the content must be clear.

### 3. Design materials

A multi-story factory building uses a reinforced concrete cast-in-place one-way slab floor system. The design data for the three floors, including floor loads, materials, and structural details, are as follows:

1 ） Reinforced concrete cast-in-situ slab,$\gamma = 25\,\mathrm{kN/m^3}$, and Plastering on the bottom of 20mm thick mixed mortar board,$\gamma = 17\,\mathrm{kN/m^3}$

2 ） The concrete strength grade is C35. Except for the longitudinal main reinforcement bars of the beams , which use HRB335 steel bars, all other steel bars are HPB300 steel bars (reinforced concrete density $\gamma = 25kN/m^3$).

3 ） The slab extends 120mm into the wall, the secondary beam extends 240mm into the wall, and the main beam extends 370mm into the wall; the column cross-section is 400 mm × 400 mm .

4) Standard value of floor live load: q = 4.5 kN/m²

# Calculation section

## I. Floor slab column grid and structural drawing design

1. The building's floor plan is rectangular, with axial dimensions of 24m × 18m. .

The slab span is generally 2.0~3m, the secondary beam span is generally 4.0~6.0m, the main beam span is 5.0~8.0m, and it should be 3 times the slab span (secondary beam).the main beams are arranged transversely, thus determining:

(1) Main girder span: 4.5m;

(2) The span of the secondary beam is: 4.4m;

(3) The span of the slab is: 1.5m.

Obtained: The floor column grid diagram is shown in Figure 1-1.

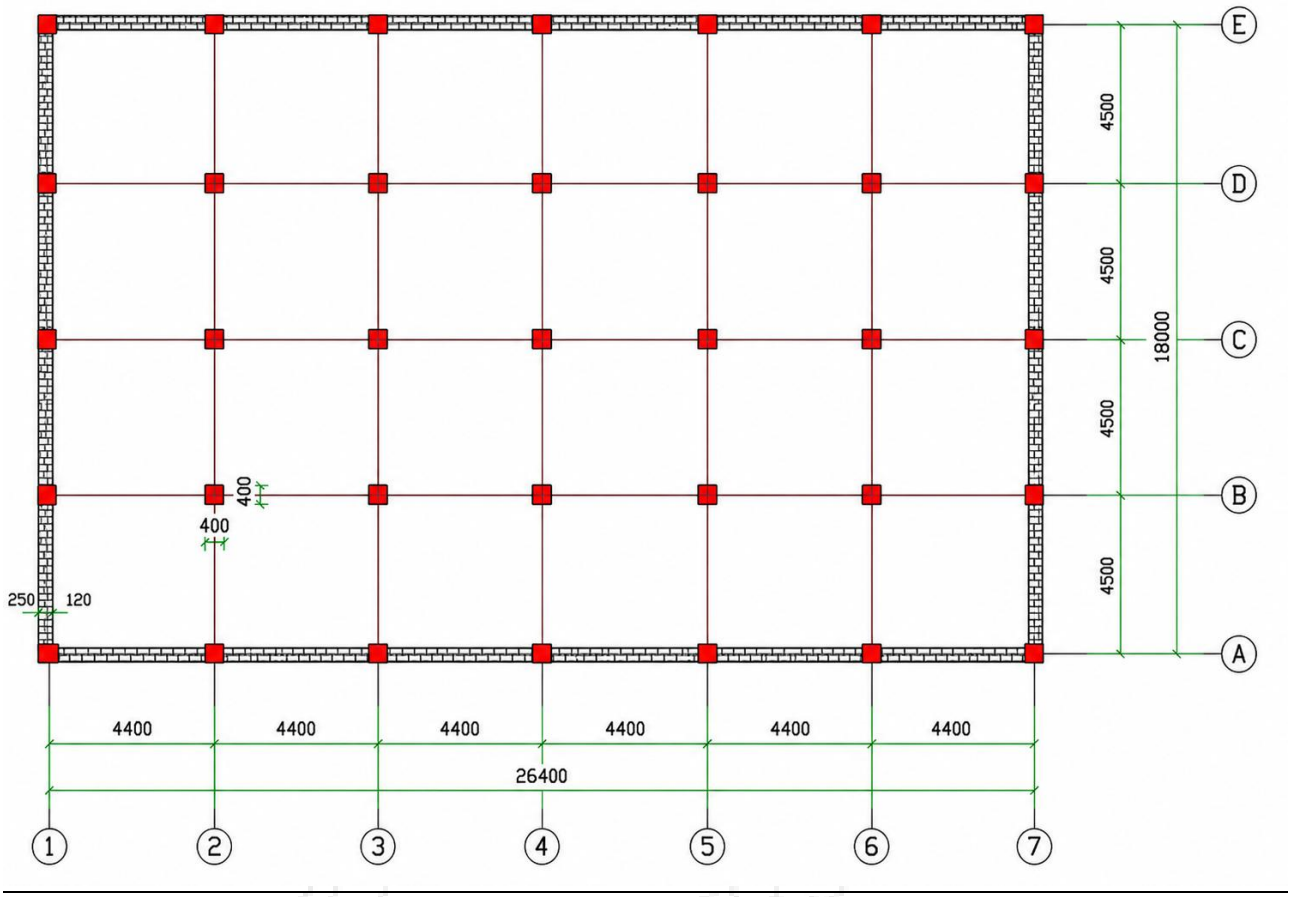



Figure 1-1: Floor column grid diagram

2. According to structural requirements, the preliminary cross-sectional dimensions are as follows:

（1）**Slab thickness**: Based on the height-to-span ratio, when $h \geq \frac{1}{30}l = \frac{1}{30} \times 1500 = 50mm$ When stiffness requirements are met, deflection can be not calculated, and the thickness requirements for industrial building floor slabs are also met $h \geq 70mm$, Take the thickness of the planks $h = 80mm$;

（2）**Secondary beam**: The cross-sectional height of the secondary beam should meet the

requirements of $h = (\frac{1}{12} \sim \frac{1}{18} L = (\frac{1}{12} \sim \frac{1}{18} \times 4400 = 322 \sim 483)mm$, Take $h = 400mm$ ;

Then $b = (\frac{1}{2} \sim \frac{1}{3})h = (133 \sim 200)mm$, Take $b = 200mm$。

（3）**Main beam** : The cross-sectional height of the main beam should meet the requirements $h = (\frac{1}{10} \sim \frac{1}{15})L = (\frac{1}{10} \sim \frac{1}{15}) \times 4500 = (360 \sim 540)mm$, $h = 500mm$, Then $h = (\frac{1}{2} \sim \frac{1}{3})h = (167 \sim 250)mm$, Take $b = 250mm$。

（4）**Column cross-section** : $b \times h = 400mm \times 400mm$

3. The beam-slab structural layout and component dimensions of the floor slab are shown in Figure 2-1

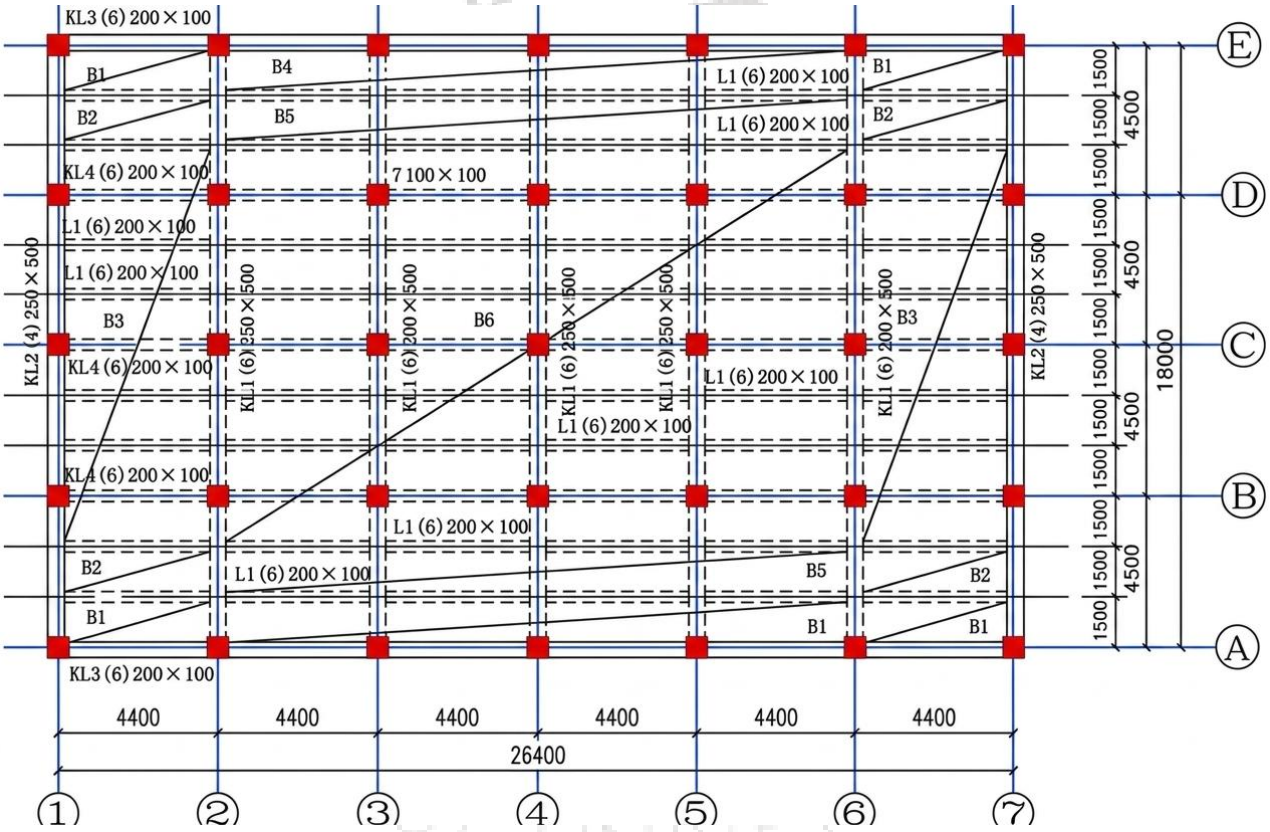



Figure 2-1 Beam-slab structural layout plan

# II. Calculation of slabs, secondary beams and main beams

## (a) Calculation of the board

### 1. Load calculation (calculated using methods considering plastic internal force redistribution)

Calculation using a 1m wide strip:

20mm thick cement mortar to the surface （$\gamma = 20kN/m^3$） $20 \times 0.02 = 0.4kN/m^2$

15mm thick lime mortar to the base ($\gamma = 17kN/m^3$) $17 \times 0.015 = 0.255kN/m^2$

80mm Thick reinforced concrete slabs ($\gamma = 25kN/m^3$) $25 \times 0.08 = 2.0kN/m^2$

Standard value of dead load： $g_k = 2.655kN/m^2$

Standard value of live load： $q_k = 5 \times 1 = 5kN/m^2$

The component factor for dead load is taken as 1.2; because the standard value of live load on the floor of an industrial building is greater than $4.0kN/m^2$, Therefore, the live load component factor is taken as 1.3.

Therefore, the total design value of the board is: $g + q$ =1.3$g_k$ +1.5$q_k$

$$1.3 \times 2.655 + 1.5 \times 5$$

$$= 10.95\ kN/m^2$$

### 2. The calculation diagram is shown in Figure 3-2

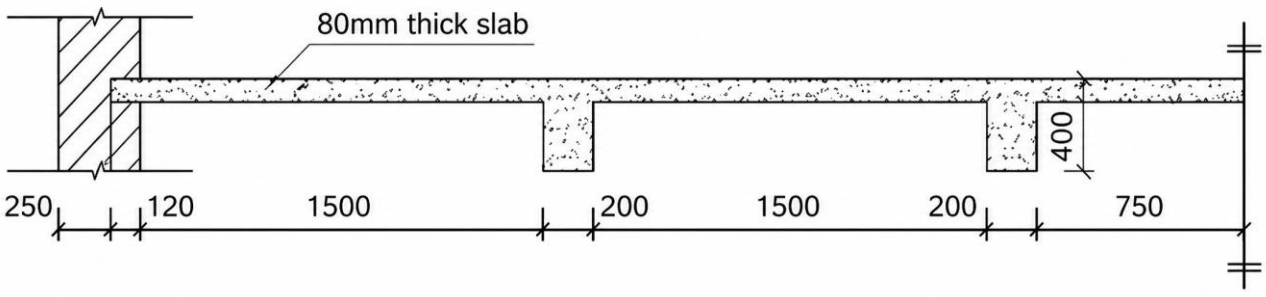



Figure 3-2 Calculation diagram of the plate

Take 1m slab width as the calculation unit, the cross-sectional dimension of the secondary beam is, $b \times h = 200mm \times 400mm$ and the slab thickness is. $80mm$

The calculation span of the board is:

Side span: $l_1 = l_0 + \frac{1}{2}h = 1500 - 100 - 120 + 80/2 = 1320mm$

Or $l_1 = l_0 + \frac{1}{2}a = 1500 - 100 - 120 + 120/2 = 1340mm$

Take the minimum value $l_1 = 1320mm \quad (a = 120mm)$

The middle span: $l_2 = l_0 = 1500 - 200 = 1300mm$

Poor span: $(1320 - 1300)/1300 = 1.54\% < 10\%$

Slabs are multi-span continuous slabs. For uniform-span continuous plates with more than five spans, if each span is equally loaded and the span difference does not exceed 10%, it can be calculated as a five-span uniform continuous slab.

Then:

The calculation diagram is shown in Figure 3-3

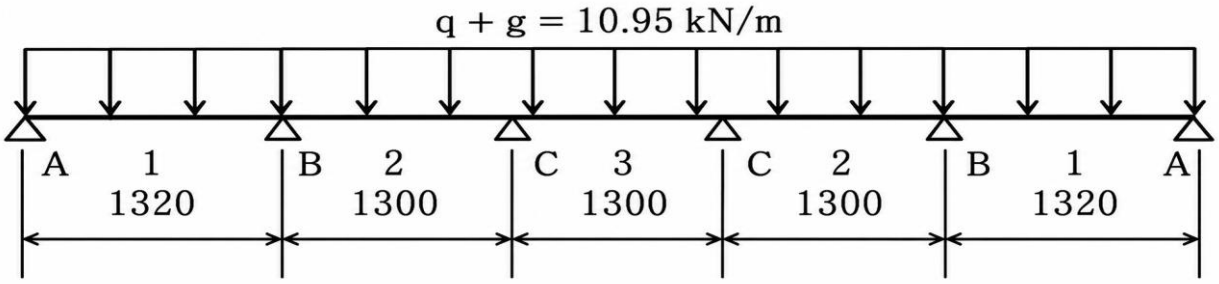



Figure 3-3 Simplified calculation diagram of the plate

3. The calculation of the internal force of the plate is shown in Table 3-1

Table 3-1

| Cross-section | 1 | B | 2 | C |
|---|---|---|---|---|
| Section coefficient α | 1/11 | -1/11 | 1/16 | -1/14 |
| $M=\alpha(g+q)l_0^2$ $(kN/m)$ | 1.734 | -1.734 | 1.157 | -1.322 |

Note: The beam slab is placed on the wall, and the moment and section calculation factor αm = 0 for support A

## 4. Calculation of cross-sectional strength

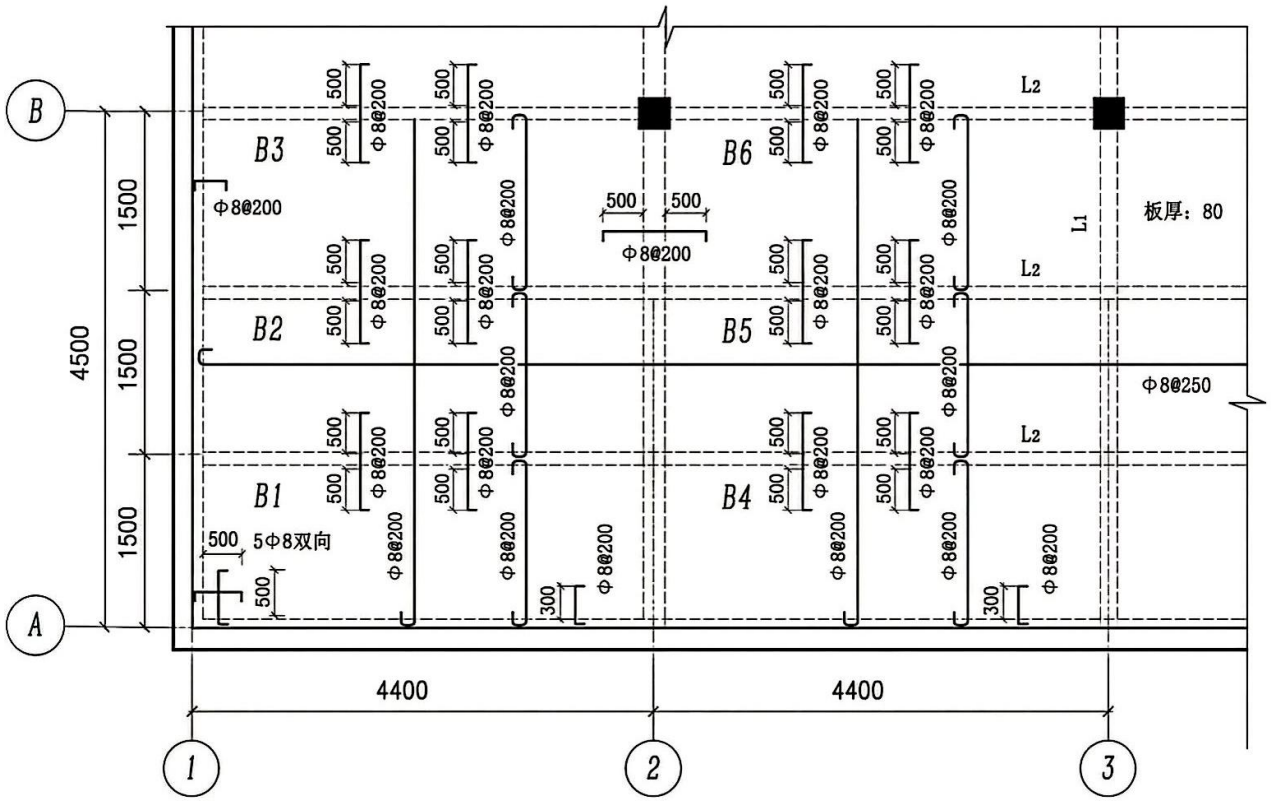
Assuming the environment category is Class I, the minimum rebar cover thickness c = 15mm is found in the table, assuming longitudinal rebar diameter d = 10mm and slab thickness 80mm, then the effective cross-sectional height as= c+d/2=20mm Take $b=1000mm$ Then $h_0=h-a_s=80-20=60mm$, Adopt C25 Coagulation ($f_c=11.9N/mm^2$), HRB335Rebar (Main Longitudinal Bar of Beam)($f_y=300N/mm^2$), Both the plate and stirrups use HPB300 rebar: ($f_y=270N/mm^2$)。Considering the favorable factors of unidirectional slab arch effects in the middle compartment connected to the beam as a whole, for the unidirectional slab of the middle compartment, the mid-span section moment of the middle span and the support section moment can each be reduced by 20%.

5）The bending moment and reinforcement calculations for the slab are as shown in 3-2as shown Table 3-2

| Cross-section | 1 | B | 2 | C |
|---|---|---|---|---|
| M（ $kN \cdot r$） | 1.734 | -1.734 | 1.157*0.8 | -1.322*0.8 |
| $\alpha_s = \frac{M}{\alpha_1 f_c b h}$ | 0.050 | -0.050 | 0.027 | -0.031 |
| $\xi = 1 - \sqrt{1-2\alpha}$ | 0.051 | 0.051 | 0.028 | 0.032 |
| $A_S = \frac{\alpha_1 f_c b \xi h}{f_y}$ | 135 | 135 | 74 | 85 |
| Rebar is selected | $\phi$8@200 | $\phi$8@200 | $\phi$6/8@200 | $\phi$6/8@200 |
| Actual reinforcement | 251.0 | 251.0 | 196.0 | 196.0 |

The support sections $\xi$ are all less than $\xi$ 0.35, which meets the principle of plastic internal force redistribution. $A_S$ / bh = 196 / (1000 * 80) = 0.245%, which is greater than: 0.45f $_t$ / f $_y$ = 0.45 * 1.27 / 270 = 0.212%, and is also greater than 0.2%, which meets the minimum reinforcement ratio requirement.

6) The reinforcement of the board is shown in Figure 3-1




Appendix 3-1 Reinforcement of the board

## (II) Calculation of secondary beams (based on plastic internal force redistribution)

### 1. Load calculation

Loads transmitted from the plates: $2.655 \times 1.5 = 3.983\ kN/m$

Next, Liang Zizhong: $0.2 \times (0.4 - 0.08) \times 25 = 1.6kN/m$

The secondary beams were heavily plastered: $0.015 \times (0.4 - 0.08) \times 2 \times 17 = 0.163kN/m$

Standard value of dead load: $g_k = 5.746kN/m$

Standard value of live load: $q_k = 5 \times 1.5 = 7.5kN/m$

Total design value: $g + q = 1.3g_k + 1.5q_k = 1.3 \times 5.746 + 1.5 \times 7.5 = 18.72kN/m$

2）Calculate the diagram 4-2 as shown

$b \times h = 250mm \times 500mm$

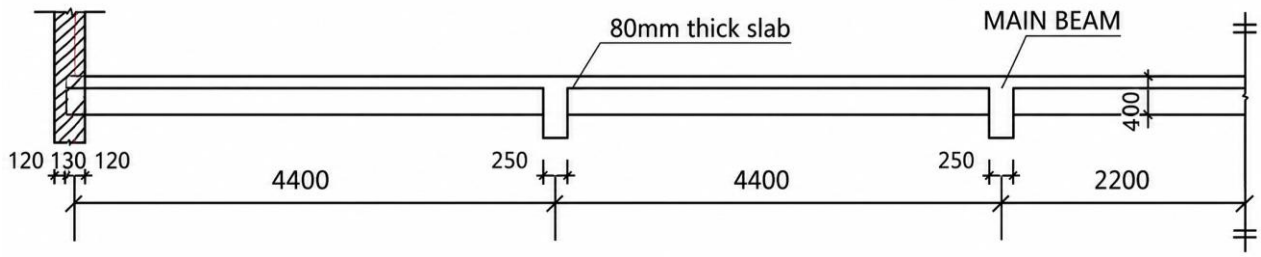



Figure 4-2 Calculation diagram of the secondary beam

Calculate span：

Side span： $l_1 = l_0 + \frac{1}{2}b = 4400 - 125 - 120 + 250/2 = 4280mm$

Or $l_1 = 1.025l_0 = 1.025 \times (4400 - 125 - 120) = 4259mm > l$

Take $l_1 = 4280mm$

The middle span： $l_2 = l_0 = 4400 - 250 = 4150mm$

Spanning difference: （4280-4150）/4150=3.13%<10%

Secondary beams are multi-span continuous beams. For continuous beams with more than five spans, if each span bears the same load and the span difference does not exceed 10%, it can be calculated as a five-span continuous beam.

Therefore , the simplified calculation diagram is shown in Figure 4-3.

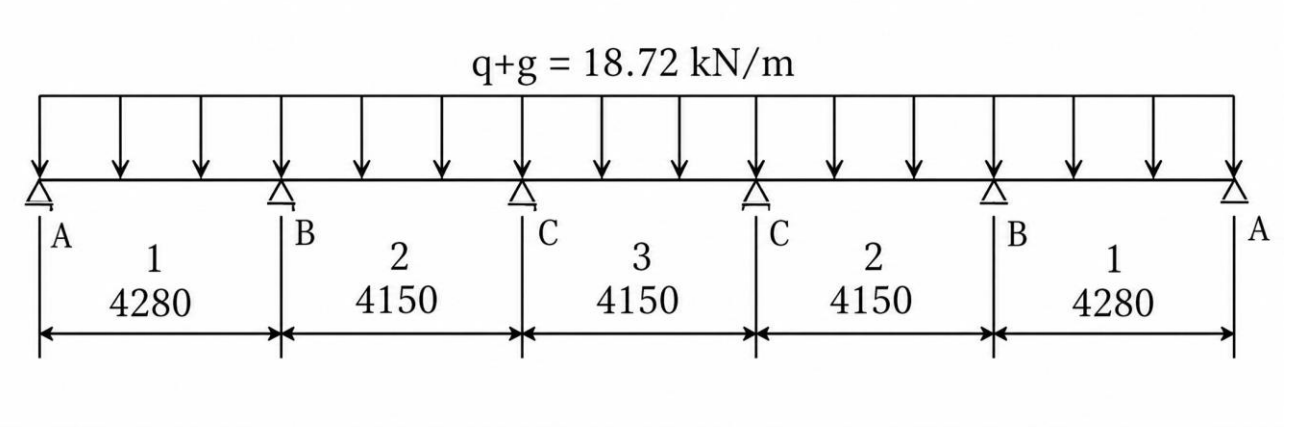



Figure 4-3 Calculation diagram of secondary beam

3）The calculation of the bending moment of the secondary beam is shown in Table 4-1.

| 截面 | 1 | B | 2 | C |
|---|---|---|---|---|
| Section modulus α | 1/11 | -1/11 | 1/16 | -1/14 |
| $M=\alpha(g+q)l_0^2$ $(kN/m)$ | 31.17 | -31.17 | 20.15 | -23.03 |

The shear force calculation for the secondary beams is shown in Table 4-2 (the clear span L0 of the beam is used for the shear force calculation ).

Table 4-2

| section | A | B (left) | B (right) | C |
|---|---|---|---|---|
| shear coefficient | 0.45 | 0.60 | 0.55 | 0.55 |
| $V=\beta(g+q)l_n$ $(kN)$ | 34.96 | 46.61 | 42.73 | 42.73 |

## DI4) Calculation of cross-sectional strength:

① Determine the flange width:

When calculating the flexural capacity of the cross-section, the section within the span is calculated as a T-section. When the secondary beam is in operation, the slab can be considered as a flange. At mid-span, the flange under compression can be calculated as a beam. The calculated flange width is taken as the smaller of the following values. Therefore, $h_f/h_0 = 80/(400-35) = 0.219 > 0.1$ only the calculated span $l$ and clear distance $S_n$ are considered. $h$ = 400 - 35 = 365mm (taking the protective layer thickness as 35mm), flange thickness... $h_f^{'} = 80mm$

**Side span:** Considered based on the calculated span $b_f^{'} \le \frac{1}{3}l = \frac{1}{3}\times 4680 = 1560mm$,

Considering the clear span of the beam (rib) $b_f^{'} = b + S_n = 200 + 4555 = 4755mm$, Pick $b_f^{'} = 1560mm$.

**Middle span:** Considered based on the calculated span $b_f^{'} \le \frac{1}{3}l = \frac{1}{3}\times 4550 = 1517mm$

Considering the clear distance between beams (ribs) $b_f^{'} = b + S_n = 200 + 4550 = 4750mm$

Pick $b_f^{'} = 1517mm$

The flange of the support section is under tension and is calculated only as a rectangular beam.

② Determine the cross-sectional type of the secondary beam

$$\alpha_1 f_c b_f' h_f' (h_0 - \frac{h_f'}{2}) = 1.0\times 11.9\times 1560\times 80\times (365-\frac{80}{2}) = 482.67\times 10^6 N\cdot mm$$

$$= 482.67kN\cdot m > \begin{cases} 53.43kN\cdot m(\quad) \\ 34.75kN\cdot m(\quad) \end{cases}$$

It belongs to the first type of T-section and $b_f^{'}$ is calculated as a rectangular section with a beam width of .

③ The calculation of the bearing capacity of the secondary beam's normal section is shown in Table 4-3.

Table 4-3 Calculation of the bearing capacity of the secondary beam's

normal section

| 截面 | 1 | B | 2 | C |
|---|---|---|---|---|
| M（$kN \cdot m$） | 31.17 | -31.17 | 20.15 | -23.03 |
| $b$或$b_f'$ | 1560 | 200 | 1517 | 200 |
| $\alpha_s = \frac{M}{\alpha_1 f_c b h_0^2}$ | 0.013 | 0.098 | 0.0084 | 0.073 |
| $\xi = 1 - \sqrt{1 - 2\alpha_s}$ | 0.013 | 0.103 | 0.0084 | 0.076 |
| $A_S = \frac{\alpha_1 f_c b \xi h_0}{f_y}$ | 327 | 332 | 215 | 290 |
| Selecting steel bars | $2\varphi12 + 1\varphi12$ | $2\varphi12 + 1\varphi12$ | $2\varphi12$ | $2\varphi12 + 1\varphi10$ |
| $A_S$ (mm) | 339 | 339 | 226 | 304.5 |
| Reinforcement ratio ρ | 0.42% | 0.42% | 0.28% | 0.38% |

All cross sections $\xi$ are less than $\xi$ 0.35 (concrete <C50, steel reinforcement HRB335), which meets the principle of plastic internal force redistribution. The reinforcement ratio ρ is greater than: $0.45f_t/f_y$ =0.45*1.27/300=0.191%<0.2%, take $\rho_{min}$=0.2%, which meets the minimum reinforcement ratio requirement.

5)The calculation of the shear section strength of the secondary beam is shown in Table 4-4.

Calculation of the strength of the inclined section of the secondary beam

Table 4-4

| section | A | B (left) | B (right) | C |
|---|---|---|---|---|
| V (kN) | 34.96 | 46.61 | 42.73 | 42.73 |
| $0.25\beta_c f_c b h_0 (kN)$ | $0.25 \times 1.0 \times 11.9 \times 200 \times 365 = 217.2kN > V$ The maximum value of 60.72 is satisfied. | | | |
| $0.7 f_t b h_0 (kN)$ | $0.7 \times 1.27 \times 200 \times 365 = 64.897kN > V$ max=61.971 Reinforcement according to structural requirements | | | |
| Number of stirrup legs and diameter | $2\varphi6$ | $2\varphi6$ | $2\varphi6$ | $2\varphi6$ |
| $A_{sv} = nA_{sv1}$ | 57 | 57 | 57 | 57 |

| | | | | |
|---|---|---|---|---|
| Actual stirrup spacing | 150 | 150 | 150 | 150 |

The reinforcement ratio $\rho = Asv / bs = 57 / (200 * 150) = 0.19\% > \rho_{sv,min} = 0.24ft / fy = 0.24 * 1.27 / 270 = 0.11\%$, which meets the minimum reinforcement ratio requirement.

## 6) Secondary beam construction

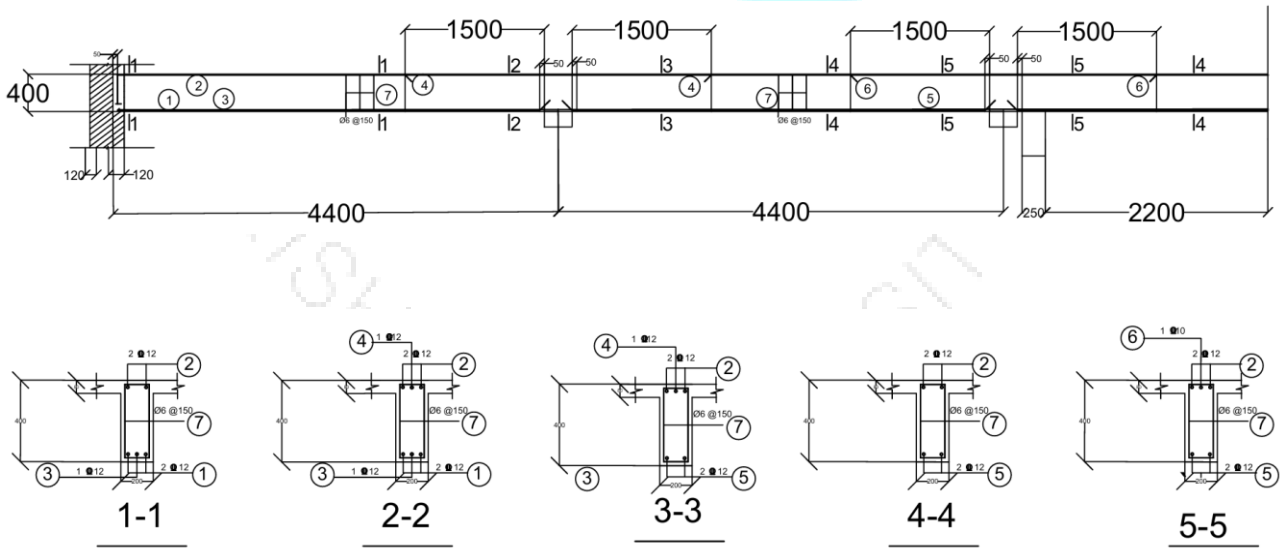
The construction of secondary beams should meet the following requirements:

① The construction of secondary beams should meet all the construction requirements for bending members;

② The length of the secondary beam extending into the wall shall not be less than 240mm;

③If the cross-sectional dimensions meet the requirements of height-to-span ratio ( $\frac{1}{18} \sim \frac{1}{12}$ ) and height-to-width ratio ( $\frac{1}{3} \sim \frac{1}{2}$ ), then deflection and crack width verification are not required.

## 7) The reinforcement diagram of the secondary beam is shown in Appendix 4-1 .




Secondary beam reinforcement diagram 4-1